

- 8 A wire is stretched elastically by a force of 200 N, causing an extension of 2.00 mm. The force is gradually increased to 250 N, and the wire remains within its elastic limit.

What is the work done in stretching the wire from 200 N to 250 N?

- A** 0.113 J **B** 0.225 J **C** 113 J **D** 225 J

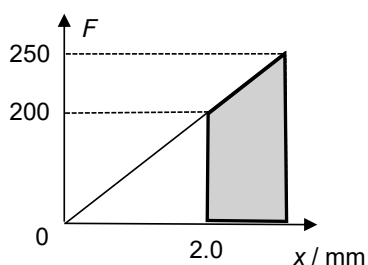
Ans: A

$$F = kx$$

$$k = \frac{200}{0.002} = 100\,000 \text{ N m}^{-1}$$

Area under graph for additional extension

$$\begin{aligned} \text{work done} &= \frac{1}{2}(200 + 250)\left(\frac{250}{100000} - 0.002\right) \\ &= 0.113 \text{ J} \end{aligned}$$



- 9 A small mass is attached to a light string and rotates in a vertical circle of radius r